Java Assessment

Instructions:

- Attempt all 100 multiple-choice questions.
- Each question has four options (A, B, C, D), with only one correct answer.
- Difficulty level: Medium to Hard.
- No answer key is provided.

Part 1: Variables, Data Types & Operators (Questions 1–30)

Q1. Which of the following declarations will cause a compilation error due to illegal variable naming?

- A. int _score = 100;
- B. int $value = 50;
- C. int 1stPlace = 10;
- D. int total_Amount = 500;

Q2. What is the default value of a static instance variable of type boolean in Java?

- A. true
- B. false
- C. null
- D. Unassigned (Compilation error)

Q3. Which of the following wrapper classes is NOT in the java.lang package by default?

- A. Integer
- B. Character
- C. BigInteger
- D. Double

Q4. What is the output of System.out.println(10 + 20 + "Java" + 10 + 20);?

- A. 30Java30
- B. 30Java1020
- C. 1020Java1020
- D. 30Java30

Q5. Which of the following primitive types has the largest memory footprint in Java?

- A. float
- B. long
- C. double
- D. Both long and double are 8 bytes (64 bits)

Q6. What happens when you compile and run: char c = 0x0041; System.out.println(c);?

- A. Prints 65
- B. Prints A
- C. Compilation error
- D. Prints 0x0041

Q7. Which of the following statements about local variables in Java is true?

- A. They are assigned default values by the JVM.
- B. They can have access modifiers like public or private.
- C. They must be initialized before they are used.
- D. They reside on the heap memory.

Q8. What is the range of values for a standard Java byte data type?

- A. -128 to 127
- B. -32768 to 32767
- C. 0 to 255
- D. -256 to 255

Q9. Which of the following floating-point literals is invalid?

- A. double d = 123.45;
- B. float f = 123.45f;
- C. double d = 123.45D;
- D. float f = 123.45;

Q10. What is the output of System.out.println(Boolean.parseBoolean("TRUE"));?

- A. true
- B. false

- C. TRUE
- D. Throws NumberFormatException

Q11. Which data type should be used to represent precise decimal values like currency in financial applications to avoid rounding errors?

- A. double
- B. float
- C. BigDecimal
- D. long representing cents

Q12. What is the result of casting (byte) 130 assuming an 8-bit signed two's complement byte?

- A. 130
- B. -126
- C. -128
- D. Compilation error

Q13. Which of the following is a valid hex literal assignment in Java?

- A. int x = 0x1A3;
- B. int x = #1A3;
- C. int x = 1A3h;
- D. int x = \u01A3;

Q14. What is the output of: char c = 'Z'; c++; System.out.println(c);?

- A. Z
- B. [
- C. A
- D. Compilation error

Q15. Which feature introduced in Java 10 allows local variable type inference?

- A. auto
- B. var
- C. let
- D. dynamic

Q16. What is the output of int x = 5; int y = ++x * x++; System.out.println(y);?

- A. 36
- B. 42
- C. 30
- D. 35

Q17. Which of the following operators has the highest precedence in Java?

- A. * (multiplication)
- B. ++ (postfix increment)
- C. + (addition)
- D. = (assignment)

Q18. What is the result of 10 >> 2 in Java?

- A. 5
- B. 2
- C. 40
- D. 2.5

Q19. What is the output of -10 >>> 2?

- A. -3
- B. 1073741821
- C. -2
- D. 0

Q20. Which expression correctly evaluates whether an integer n is even without using the modulus operator?

- A. (n & 1) == 0
- B. (n ^ 1) == 0
- C. (n | 1) == 0
- D. n >> 1 == n / 2

Q21. What is the result of true || false && false?

- A. false

- B. true
- C. Compilation error
- D. NullPointerException

Q22. What is the output of System.out.println(10 == 10.0);?

- A. false
- B. true
- C. Compilation error
- D. Runtime exception

Q23. Which of the following assignments will cause a compile-time error?

- A. byte b = 10; b += 5;
- B. byte b = 10; b = b + 5;
- C. short s = 20; s++;
- D. char c = 'a'; c += 3;

Q24. What is the output of int a = 5; int b = 3; System.out.println(a & b);?

- A. 7
- B. 1
- C. 5
- D. 3

Q25. What is the output of String s = null; System.out.println(s instanceof String);?

- A. true
- B. false
- C. NullPointerException
- D. Compilation error

Q26. What does the ternary expression int x = (5 > 3) ? 10 : 20.5; cause?

- A. Evaluates to 10
- B. Evaluates to 20.5
- C. Compile-time error due to type mismatch in branches
- D. Implicitly casts 10 to 10.0

Q27. What is the output of System.out.println(5 ^ 3);?

- A. 6
- B. 5
- C. 3
- D. 0

Q28. Which operator is used for unsigned right shift in Java?

- A. >>
- B. >>>
- C. <<
- D. >~

Q29. What is the result of double d = 0.1 + 0.2; System.out.println(d == 0.3);?

- A. true
- B. false
- C. Compile-time error
- D. Runtime exception

Q30. What is the output of: int x = 10; System.out.println(~x);?

- A. -10
- B. -11
- C. 9
- D. 10

Part 2: Control Flow (If-Else & Switch-Case) (Questions 31–55)

Q31. Which of the following data types CANNOT be used in a switch expression control variable in Java?

- A. String
- B. int
- C. double
- D. char

Q32. What is the output of the following code snippet?

Java

```
int x = 10;
if (x > 5)
   if (x > 15)
      System.out.print("A ");
   else
      System.out.print("B ");
else
   System.out.print("C ");
```

- A. A
- B. B
- C. C
- D. No output

Q33. In a Java switch statement, what happens if a break statement is omitted at the end of a case block?

- A. Compilation error
- B. Execution falls through to the next case automatically
- C. The program terminates immediately
- D. It skips the default case

Q34. Which of the following is valid syntax for a Java 14+ switch expression?

- A. case 1 -> System.out.println("One");
- B. case 1 : -> System.out.println("One");
- C. case 1 => System.out.println("One");
- D. case 1 : yield "One";

Q35. What is the output of:

Java

```
boolean flag = false;
if (flag = true) {
```

```
    System.out.println("True");
} else {
    System.out.println("False");
}
```

- A. True
- B. False
- C. Compilation error
- D. Runtime exception

Q36. What is the output of the following switch snippet?

Java

```
int val = 2;
switch(val) {
    case 1: System.out.print("1 ");
    case 2: System.out.print("2 ");
    case 3: System.out.print("3 ");
    default: System.out.print("D ");
}
```

- A. 2
- B. 2 3 D
- C. 2 3
- D. D

Q37. Can a switch statement expression evaluate to null in Java?

- A. Yes, without any issues.
- B. No, it throws a NullPointerException at runtime if the expression evaluates to null.
- C. No, it is a compile-time error.
- D. Yes, it automatically executes the default block.

Q38. What happens if you attempt to access a local variable declared inside an if block outside of that block?

- A. Prints default value
- B. Returns null
- C. Compile-time error due to variable scope
- D. Runtime exception

Q39. Which keyword was introduced in Java 13 to return a value from a switch expression block?

- A. return
- B. yield
- C. break
- D. value

Q40. What is the output of:

Java

```
int score = 85;
if (score >= 90) System.out.print("A");
else if (score >= 80) System.out.print("B");
else if (score >= 85) System.out.print("C");
```

- A. A
- B. B
- C. C
- D. B C

Q41. Can a float or long be used as a switch selector expression in Java?

- A. Both can.
- B. Neither can.
- C. float can, long cannot.
- D. long can, float cannot.

Q42. What is the output of:

Java

```
int i = 1;
```

```
switch (i) {
   case 1: System.out.print("One "); break;
   case 1: System.out.print("Duplicate "); break;
   default: System.out.print("Default ");
}
```

- A. One Duplicate
- B. One
- C. Compile-time error (duplicate case label)
- D. Runtime exception

Q43. Which of the following is true regarding nested if-else statements?

- A. An else is always paired with the nearest preceding if that doesn't have an else.
- B. An else is paired with the outermost if.
- C. Curly braces are strictly required for nested if-else.
- D. Nested if-else is prohibited in modern Java.

Q44. What is the output of:

Java

```
int x = 10;
if (x < 5); {
   System.out.println("Hello");
}
```

- A. Hello
- B. No output
- C. Compile-time error
- D. Runtime exception

Q45. In a Java switch case, can case labels be expressions containing variables?

- A. Yes, any variable can be used.
- B. No, case labels must be compile-time constants (or constant expressions).
- C. Only final local variables can be used.

- D. Only enum constants can be used.

Q46. What is the output of:

Java

```
String day = "MONDAY";
String result = switch (day) {
   case "MONDAY", "FRIDAY", "SUNDAY" -> "Active";
   default -> "Inactive";
};
System.out.println(result);
```

- A. Inactive
- B. Active
- C. Compile-time error
- D. MONDAY

Q47. What happens if no case matches the switch expression and there is no default case?

- A. Compilation error
- B. NullPointerException is thrown
- C. The switch statement completes silently without executing any code inside it
- D. It repeats the switch check

Q48. Which statement is correct about enums in switch cases?

- A. You must prefix case labels with the enum class name (e.g., case Day.MONDAY).
- B. You must NOT prefix case labels with the enum class name (e.g., use case MONDAY).
- C. Enums cannot be used in switch statements.
- D. Enums require a default case always.

Q49. What is the output of:

Java

```
int a = 5, b = 10;
if (a > b)
```

```
    System.out.println("A");
else
    System.out.println("B");
    System.out.println("C");
```

- A. B followed by C
- B. B
- C. C
- D. Compile-time error

Q50. Can a switch statement contain another switch statement?

- A. No, nesting switches is illegal.
- B. Yes, switches can be nested arbitrarily.
- C. Only if the inner switch uses different data types.
- D. Only inside the default block.

Q51. What is the result of compiling a switch expression that does not cover all possible values of an enum?

- A. It compiles successfully.
- B. It produces a compile-time error unless a default branch is provided.
- C. It throws a MatchException at runtime.
- D. It defaults to the first enum value.

Q52. Which block in an if-else if-else chain is mandatory?

- A. Only the if block.
- B. All else if blocks.
- C. The else block.
- D. Both if and else blocks.

Q53. What is the output of:

Java

```
int x = 5;
if (x == 5)
```

```
   x = 10;
else
   x = 15;
System.out.println(x);
```

- A. 5
- B. 10
- C. 15
- D. Compile-time error

Q54. In a Java switch statement, can case constants be null?

- A. Yes, starting from Java 21 (pattern matching for switch allows null cases).
- B. No, never.
- C. Yes, in all Java versions.
- D. Only for primitive types.

Q55. What is the output of:

```
Java
int x = 3;
switch (x) {
   case 1: System.out.println("One");
   case 2: System.out.println("Two");
   default: System.out.println("Default");
}
```

- A. Default
- B. One Two Default
- C. Compile-time error
- D. No output

Part 3: Loops (For, While, Do-While) (Questions 56–80)

Q56. What is the output of the following code snippet?

```
Java
```

```
int count = 0;
for (int i = 0; i < 3; i++) {
   for (int j = 0; j < 3; j++) {
      if (i == j) break;
      count++;
   }
}
System.out.println(count);
```

- A. 3
- B. 6
- C. 0
- D. 9

Q57. Which of the following for-each loops is valid for iterating over an integer array arr?

- A. for (int i : arr)
- B. for (int arr : i)
- C. for (i : int arr)
- D. for each (int i in arr)

Q58. How many times will the following do-while loop execute?

Java

```
int x = 5;
do {
   System.out.println(x);
   x--;
} while (x > 5);
```

- A. 0 times
- B. 1 time
- C. 5 times
- D. Infinite times

Q59. What is the output of:

Java

```
int i = 0;
for (System.out.print("A"); i < 2; System.out.print("B"), i++) {
   System.out.print("C");
}
```

- A. ACBCB
- B. ACBC
- C. ABCBC
- D. ACAB

Q60. What is the purpose of a labeled break statement in Java loops?

- A. To break out of the innermost loop only.
- B. To break out of an outer nested loop from an inner loop.
- C. To restart a loop from the beginning.
- D. To skip the current iteration and proceed to the next.

Q61. What is the output of:

Java

```
int sum = 0;
for (int i = 1; i <= 5; i++) {
   if (i == 3) continue;
   sum += i;
}
System.out.println(sum);
```

- A. 15
- B. 12
- C. 9
- D. 10

Q62. What happens if the boolean condition in a while loop is initialized to false?

- A. Compile-time error
- B. Loop body executes once
- C. Loop body never executes
- D. Runtime exception

Q63. Which loop construct is guaranteed to execute its block of statements at least once?

- A. for loop
- B. while loop
- C. do-while loop
- D. Enhanced for-each loop

Q64. What is the output of:

Java

```
for (int i = 0; i < 10; i++) {
   if (i % 2 == 0) continue;
   System.out.print(i + " ");
}
```

- A. 0 2 4 6 8
- B. 1 3 5 7 9
- C. 0 1 2 3 4 5 6 7 8 9
- D. 2 4 6 8

Q65. Can you have an infinite for loop in Java?

- A. Yes, e.g., for (;;)
- B. No, Java for loops must have a termination condition.
- C. Only if using while.
- D. Yes, but it requires a break statement to compile.

Q66. What is the scope of a variable declared inside the initialization section of a for loop?

- A. Accessible anywhere in the method.
- B. Accessible only within the body and header of that for loop.
- C. Accessible across all nested loops in the class.

- D. Accessible only after the loop completes.

Q67. What is the output of:

Java

```
int i = 5;
while (--i > 0) {
   System.out.print(i + " ");
}
```

- A. 5 4 3 2 1
- B. 4 3 2 1
- C. 5 4 3 2
- D. 4 3 2 1 0

Q68. Which of the following statements about continue is correct?

- A. It terminates the entire loop execution.
- B. It skips the remaining statements in the current iteration and proceeds to the loop increment/condition check.
- C. It can only be used inside switch statements.
- D. It causes a compilation error if used in a while loop.

Q69. What is the output of:

Java

```
int x = 1;
for (; x <= 3; x++) {
   System.out.print(x);
}
```

- A. 123
- B. 1234
- C. 12
- D. Compile-time error

Q70. Can multiple variables of different types be initialized in the initialization clause of a standard for loop?

- A. Yes, using commas.
- B. No, only variables of the exact same type can be declared and initialized together.
- C. Yes, without any restrictions.
- D. No, for loops only support a single variable.

Q71. What is the output of:

Java

```
int i = 0;
do {
   i++;
   if (i == 2) continue;
   System.out.print(i + " ");
} while (i < 4);
```

- A. 1 3 4
- B. 2 3 4
- C. 1 2 3 4
- D. 1 3

Q72. What is the output of:

Java

```
outer: for (int i = 1; i <= 2; i++) {
   inner: for (int j = 1; j <= 2; j++) {
      if (i == 1 && j == 1) continue outer;
      System.out.print(i + "" + j + " ");
   }
}
```

- A. 11 12 21 22
- B. 21 22
- C. 12 21 22
- D. 11 21

Q73. Which of the following loop structures results in a compile-time error due to unreachable code?

- A. while(true) { ... } followed by System.out.println("Done");
- B. for(;;) { ... } followed by System.out.println("Done");
- C. while(false) { System.out.println("Hello"); }
- D. do { ... } while(false);

Q74. What is the output of:

Java

```
int[] nums = {10, 20, 30};
for (int n : nums) {
   n = 0;
}
System.out.println(nums[0]);
```

- A. 0
- B. 10
- C. 30
- D. Compile-time error

Q75. What is the output of:

Java

```
int i = 0;
while (i < 3) {
   System.out.print(i++);
}
```

- A. 012
- B. 123
- C. 0123
- D. 12

Q76. Which keyword is required when writing an infinite loop using a while statement?

- A. true
- B. infinite
- C. loop
- D. forever

Q77. What is the output of:

Java

```
for (int i = 0; i < 3; i++) {
   int x = i;
   System.out.print(x + " ");
}
```

- A. 0 1 2
- B. 0 0 0
- C. 2 2 2
- D. Compile-time error

Q78. Can the condition expression of a for loop be omitted?

- A. No, it is mandatory.
- B. Yes, it defaults to true.
- C. Yes, but it causes a runtime exception.
- D. Only in enhanced for-each loops.

Q79. What is the output of:

Java

```
int x = 0;
for (int i = 0; i < 5; i++) {
   x += i;
}
System.out.println(x);
```

- A. 10
- B. 15

- C. 5
- D. 0

Q80. What happens if a break statement is used outside of any loop or switch statement in Java?

- A. It terminates the program.
- B. It acts as a no-op.
- C. It causes a compile-time error.
- D. It returns from the enclosing method.

Part 4: Methods (Questions 81–100)

Q81. Which of the following method signatures represents a valid varargs method in Java?

- A. public void test(int... nums, String s)
- B. public void test(String s, int... nums)
- C. public void test(int... nums1, String... nums2)
- D. public void test(int nums...)

Q82. What is method overloading in Java?

- A. Defining multiple methods with the same name and same parameter list in different classes.
- B. Defining multiple methods with the same name but different parameter lists within the same class.
- C. Redefining a superclass method in a subclass with the same signature.
- D. Calling a method recursively.

Q83. What is the output of the recursive method call print(3)?

Java

```
static void print(int n) {
    if (n == 0) return;
    System.out.print(n + " ");
    print(n - 1);
    System.out.print(n + " ");
}
```

- A. 3 2 1 1 2 3
- B. 3 2 1 3 2 1
- C. 1 2 3 3 2 1
- D. 3 2 1

Q84. Which of the following modifiers cannot be applied to an abstract method?

- A. public
- B. protected
- C. private
- D. abstract

Q85. Can a method in Java return multiple values using a standard return statement?

- A. Yes, separated by commas.
- B. Yes, using an array or object wrapper.
- C. No, Java methods can only return a single value (or void).
- D. Yes, using tuple syntax.

Q86. What is method hiding in Java?

- A. Overriding a static method in a subclass.
- B. Declaring a method as private.
- C. Defining a method inside an interface.
- D. Using anonymous inner classes.

Q87. What is the output of:

Java

```
class Test {
   static void change(int x) {
      x = 10;
   }
   public static void main(String[] args) {
      int x = 5;
      change(x);
```

```
        System.out.println(x);
    }
}
```

- A. 10
- B. 5
- C. Compile-time error
- D. 0

Q88. Which of the following is true regarding method return type during method overloading?

- A. Overloaded methods must have different return types.
- B. Overloaded methods can have the same or different return types, provided the parameter lists differ.
- C. Overloaded methods must have void return types.
- D. Return type is part of the method signature in Java.

Q89. What happens if a recursive method in Java lacks a proper base case?

- A. It compiles successfully and executes infinite iterations until StackOverflowError occurs.
- B. Compile-time error.
- C. It executes once and returns null.
- D. It throws OutOfMemoryError.

Q90. What is the access modifier of a method if no modifier is explicitly specified?

- A. public
- B. private
- C. protected
- D. Package-private (default)

Q91. Can a static method access non-static (instance) variables directly without an object reference?

- A. Yes, always.
- B. No, it results in a compile-time error.

- C. Only if the instance variables are final.
- D. Only within the main method.

Q92. What is the output of:

Java

```
class Test {
   static int add(int a, int b) { return a + b; }
   static double add(double a, double b) { return a + b; }
   public static void main(String[] args) {
      System.out.println(add(5, 10));
   }
}
```

- A. 15
- B. 15.0
- C. Compile-time error (ambiguous method call)
- D. Runtime exception

Q93. Which keyword is used to define a method that does not return any value?

- A. null
- B. empty
- C. void
- D. nil

Q94. Can a method be declared as both abstract and final in Java?

- A. Yes.
- B. No, they are mutually exclusive (abstract requires overriding, final prevents it).
- C. Only in interfaces.
- D. Only in abstract classes.

Q95. What is the signature of the standard main method in Java?

- A. public static void main(String[] args)
- B. public void main(String args[])

- C. static public int main(String[] args)
- D. private static void main(String... args)

Q96. What is a "pure" or deterministic method in the context of functional programming support in Java?

- A. A method that has no side effects and always returns the same result for the same arguments.
- B. A method declared with the pure keyword.
- C. A method that only modifies static fields.
- D. A method that returns void.

Q97. How does Java pass arguments to methods?

- A. Pass by reference for objects and pass by value for primitives.
- B. Pass by value for all arguments (references to objects are passed by value).
- C. Pass by reference for all arguments.
- D. Pass by pointer.

Q98. What is the output of:

Java

```
class Test {
   static void test(int... a) {
      System.out.print("Varargs ");
   }
   static void test(int a, int b) {
      System.out.print("Fixed ");
   }
   public static void main(String[] args) {
      test(1, 2);
   }
}
```

- A. Varargs
- B. Fixed

- C. Compile-time error
- D. Fixed Varargs

Q99. Can a static method be overridden in a subclass?

- A. Yes, using the @Override annotation.
- B. No, static methods cannot be overridden (they are hidden, not overridden).
- C. Yes, automatically.
- D. Only if declared final.

Q100. What is the maximum number of values a Java method can return directly using a return statement?

- A. 1
- B. Unlimited if wrapped in a collection or object
- C. As many as specified in the throws clause
- D. 2 (using pair objects)